DAEN 300 Data Engineering Coding Experience I

Fall 2026

Course Information

Time (Location) Fri 12:40–3:30 PM *0 Lecture Hours. 3 Lab Hours.* (ZACH 344)
Prerequisites Grade of C or better in ECEN 360; concurrent enrollment in DAEN 321 and DAEN 331; junior or senior.
Instructor Alexander Abuabara `mailto:abuabara@tamu.edu`
Course office hours (location) Wed 1-3 PM or by appointment (ETB 4013)

Description

Application of computational tools to model and solve data engineering problems primarily involving machine learning and optimization techniques.

Outcomes

1. Use Python, Jupyter Notebooks, and widely used Python libraries to solve data engineering problems.
2. Work with common data formats such as CSV and JSON.
3. Clean, transform, and analyze data using NumPy, pandas, Matplotlib, and Scikit-Learn.
4. Apply basic algorithms, including searching, sorting, and recursion.
5. Build and evaluate basic machine learning models.
6. Improve computational solutions using basic optimization techniques.
7. Create and explain an end-to-end data engineering and analysis workflow.

Recommended Resources

- Python Data Science Handbook (PDSH) `https://jakevdp.github.io/PythonDataScienceHandbook/`.
- The Recursive Book of Recursion (RBR) `https://inventwithpython.com/recursion/`.

Grading

Assessment	Weight	Description
Weekly Assignments (12–13)	89%	Weekly activities and worksheets; each weighted equally.
In-Class Participation	11%	In-class quizzes, discussions, participation

Grades are based on total points earned: A ≥ 90%, B ≥ 80%, C ≥ 70%, D ≥ 60%, F < 60%.

Schedule (*tentative*)

Week	Topic	Reading	Lab/Coding
1	Python and Jupyter: functions, debugging, profiling	*PDSH* 1, *RBR* 1	Load and explore a dataset in Jupyter
2	NumPy: arrays, indexing, vectorization, aggregation	*PDSH* 2	Process numerical data
3	(pandas) Dataframes, filtering, missing data, joins	*PDSH* 3	Clean and combine datasets
4	Data transformation: grouping, pivot tables, strings	*PDSH* 3	Summarize and transform data
5	Recursive + iterative solutions, base cases, call stacks	*RBR* 1+2	Compare recursive and iterative programs
6	Algorithms: binary search, merge sort, quick sort	*RBR* 3+5	Implement searching and sorting algorithms
7	Matplotlib: visualization and exploratory analysis	*PDSH* 4	Explore and visualize a dataset
8	Scikit-Learn: machine learning basics	*PDSH* 5	Build a basic machine learning pipeline
9	Linear regression	*PDSH* 5	Build, evaluate a regression model
10	Classification, model evaluation	*PDSH* 5	Train and compare classification models
11	Decision trees, random forests, tree traversal	*PDSH* 5, *RBR* 4	Build tree-based models
12	Clustering, PCA, model selection	*PDSH* 5	Apply clustering, dimensionality reduction
13	Optimization: backtracking, dynamic programming	*RBR* 6-7-8	Solve, improve an optimization problem
14	Mini-project: data prep → ML → optimization	–	An end-to-end data engineering solution

Policies

- Be nice. Be honest. Don't cheat. Adhere to all University Policies.
- Course material is cumulative and requires consistent effort. All assignments must be completed carefully and on time. Presentation quality (organization, clarity, tables, graphs, and discussion) is part of grading.
- Regular attendance and active participation are essential to successful completion of the course.
- Excused absences apply only to attendance and in-class activities and do not extend assignment deadlines. Alternative arrangements must be made prior to the deadline.
- Make-up assignments may be given orally upon the student's return from a documented, verifiable absence, on a date set by the professor.
- All assignments must be submitted through Canvas. If technical issues prevent submission, send your assignment from your `@tamu.edu` email to the instructor's `@tamu.edu` email as a backup. The subject line must include the course number and assignment name. Late work is not accepted and will receive a grade of zero.

- Regrade applies to the entire submission. To ensure fairness and timely feedback, no changes after one week.
- To provide some flexibility during the semester, you may miss one assignment without providing a university-approved excuse, and that assignment will be dropped. Students who complete and submit every assignment will receive an additional benefit: their two lowest assignment grades will be dropped.
- Audio or video recording of lectures or any portion of the course is prohibited without written instructor permission. Unauthorized recording constitutes a violation of student conduct.
- Headphones and non-class-related browsing are not permitted during class.
- Students may use AI tools to support brainstorming, research, organization, revision, and problem-solving. However, students remain fully responsible for the accuracy, quality, and integrity of submitted work and must verify AI-generated information. All outside sources, including substantial AI contributions, should be cited or acknowledged when appropriate. AI should support—not replace—your own thinking, judgment, and understanding. Students may be asked to explain orally or demonstrate the reasoning behind their work. **Use AI only in ways you can confidently and openly acknowledge.**
- Use of others' ideas is encouraged as part of scholarly work, but all sources must be properly cited. Proper attribution is required for all coursework and projects.